[MP2] Internal fragmentation optimization problem: free cache slab

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秦驊佑 (CHIN, HUA-YU)

All Sections

Assume that internal fragmentation optimization is implemented, support in cache objects = 7, max objects per slab = 8, MP2_MIN_AVAIL_SLAB = 1.
After allocating memory for 8 objects, the cache slab should be full and a new slab is allocated.
Now, free the first 7 objects, which are all stored in the cache memory.
This makes the cache slab empty and there are 2 "available slabs" now, which doesn't satisfy the memory saving condition.
How should we address this issue?

Here are some ideas (though may not be correct):

1. Move the object(s) in the newest allocated slab to the cache slab and free the slab.
2. Don't count the cache slab when counting available slab
3. Free the cache slab (which is probably not the right answer)
4. Don't consider cache slab a slab anymore and implement just as internal fragmentation optimization is not done.

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謝典樺 (HSIEH, TIEN-HUA) (https://cool.ntu.edu.tw/courses/46296/users/147125)

Friday

Available slabs count only partial and free slabs, so I think your second idea is correct. Even if cache slab is empty, it does not count as an available slab.

Reply

李承瑜 (LI,CHENG-YU) (https://cool.ntu.edu.tw/courses/46296/users/183795)

Saturday

想請問助教如果有實作kmem_cache 的內部碎裂，也就是使用cache 部分內容存obj時。 如果遇到在執行kmem_cache_free時同時遇到這三種情況

1. cache slab become empty
2. slab（即 partial + free）的總數超過 param.h 中定義的MP2_MIN_AVAIL_SLAB
3. free slab list為空

這時候函式的行為應該如何

進入

畢竟理論上它應該要free 一個slab，這時候我還應該輸出

[SLAB] slab () is freed due to save memory

嗎

Reply

(http

楊子慶 (YANG, TZU-CHING) (https://cool.ntu.edu.tw/courses/46296/users/229158)

Sunday

李同學您好：

對於 in-cache fragmentation 的實作，若釋放 in-cache object，不需用進行進一步的 slab 釋放檢查流程。

這是由於不應該將 in-cache slab (a.k.a. kmem_cache) 進行釋放。

子慶

Edited by 楊子慶 (YANG, TZU-CHING) (https://cool.ntu.edu.tw/courses/46296/users/229158) on Mar 30 at 12:42am

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